LLMOps Assignment 2: API Cost, Latency and Prompt Version Management

Course: MLOps | **Scenario:** EduAssist AI academic assistant
Model: Qwen 2.5 1.5B (qwen2.5:1.5b)
Environment: Python 3.13.9, Ollama 0.40.14, MLflow 3.18.0

1. Objective and Setup

The objective was to evaluate three academic question-answering prompts. MLOps manages the model lifecycle; LLMOps also versions prompts and monitors tokens, latency, cost and output quality.

OpenAI API access was unavailable, so the assignment's local-model alternative was used. Qwen 2.5 1.5B ran through Ollama's REST endpoint, POST /api/generate; MLflow used SQLite locally.

Five academic questions were tested against each prompt, producing 15 measured requests with the same model and generation settings. Warm-up and preliminary validation calls were excluded.

2. Prompt Evaluation and Latency

The prompts were registered under eduassist_academic_qa in the MLflow experiment eduassist_llmops_assignment. Version 1 used a basic instruction; Version 2 added academic context and concise undergraduate-level guidance; Version 3 required JSON containing answer, confidence and key_points.

For each request, the notebook recorded question ID, prompt version, model, timestamps, latency, status and token counts. Ollama's prompt_eval_count and eval_count supplied token counts. JSON validation checked parsing, required keys and types.

All 15 requests returned HTTP 200. Average latency was 15.9575 seconds for v1, 15.9443 seconds for v2 and 13.8062 seconds for v3. Overall mean latency was 15.2360 seconds, median 13.6961 seconds, minimum 5.9820 seconds, maximum 29.8436 seconds and standard deviation 6.8982 seconds.

The slowest request was Q2 with v3 at 29.8436 seconds. It generated 512 tokens and stopped with done_reason=length, reaching the configured output limit. The v1 response for Q4 also reached this limit. Longer generation and local model/hardware processing may contribute to latency.

3. Token Consumption and Cost

Across the 15 requests, the experiment used 1,086 input tokens and 3,735 output tokens. Hosted-cost estimates used the Featherless Qwen2.5-1.5B-Instruct reference rates on 5 November 2026: USD 0.04 per million input tokens and USD 0.08 per million output tokens. Input and output costs were calculated separately and projected to 10,000 requests.

Estimated monthly hosted costs were USD 0.22832 for v1, USD 0.23728 for v2 and USD 0.21888 for v3. These are hosted-inference estimates, not actual charges; local Ollama inference had no per-request cloud API charge. A cost-optimisation recommendation is to limit unnecessary response length while preserving useful academic detail.

4. Prompt Comparison and Production Promotion

Prompt version	Average latency (s)	Average input tokens	Average output tokens	Average cost/request (USD)	Valid JSON rate	Recommended?
v1 Basic	15.9575	46.4	262.2	0.000022832	0%	No
v2 Structured	15.9443	68.4	262.4	0.000023728	0%	No
v3 JSON	13.8062	102.4	222.4	0.000021888	80%	Yes

Versions 1 and 2 achieved 0% JSON validity. Version 3 achieved 80% (four of five responses). Three MLflow runs logged avg_latency_seconds, avg_input_tokens, avg_output_tokens, avg_cost_per_request and valid_json. Prompt versions and differences were reviewed, and production was assigned to Version 3.

Promotion justification. Version 3 is recommended because it best meets the application's structured-output requirement and has the strongest measured operational results. Its JSON validity was 80%, compared with 0% for v1 and v2. Its average latency, 13.8062 seconds, was lower than v1 (15.9575 seconds) and v2 (15.9443 seconds). It also generated fewer output tokens on average (222.4 versus 262.2 and 262.4), giving it the lowest estimated hosted cost per request, USD 0.000021888, and the lowest 10,000-request projection, USD 0.21888. Validation checked the required JSON structure, but factual accuracy was not independently scored; academic correctness should be reviewed before deployment. This structured response format can support downstream systems that require machine-readable answers. One v3 response (Q2) reached the 512-token limit and failed JSON validation. Version 3 was promoted for this assignment, with truncation handling, schema validation and retry or repair logic recommended before real production use. This is a controlled assignment promotion, not evidence of unconditional production readiness.

5. Conclusion

The experiment demonstrated local inference, latency and token measurement, cost estimation, prompt registration, evaluation and alias-based promotion in MLflow.